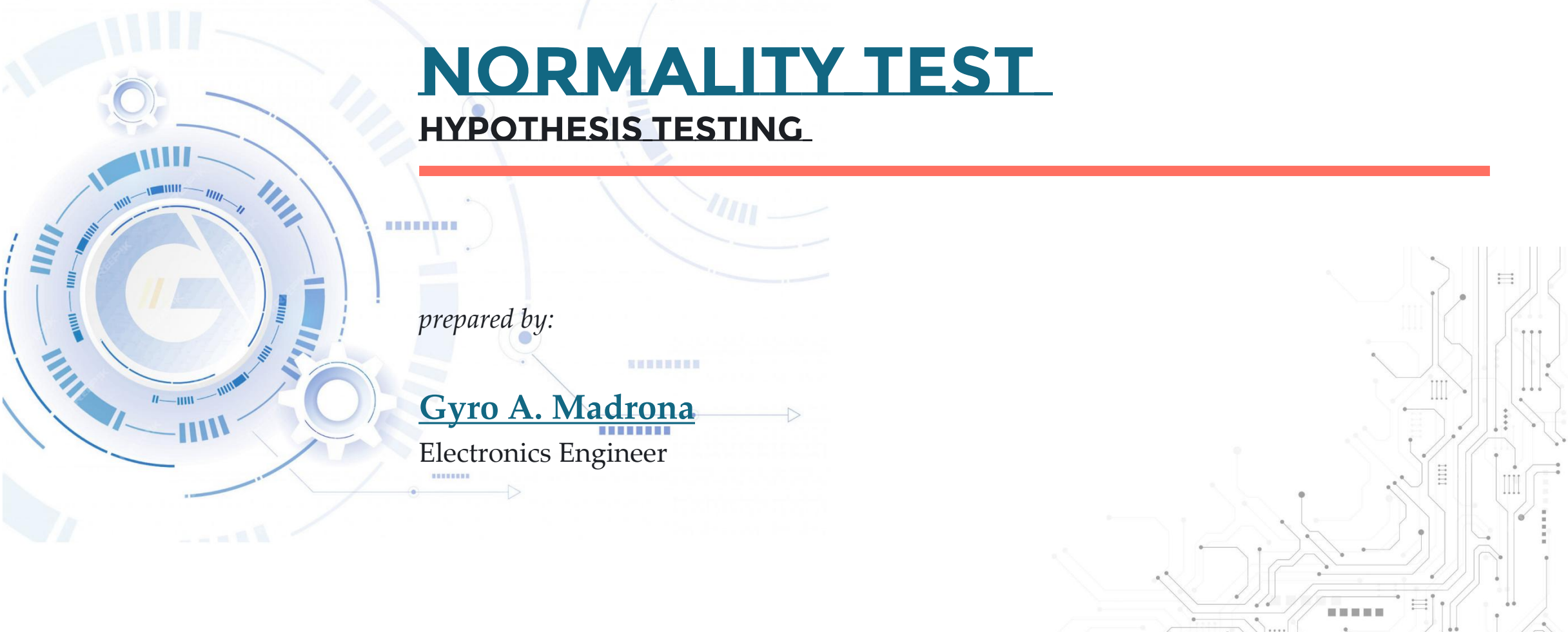




NORMALITY TEST

HYPOTHESIS TESTING

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TOPIC OUTLINE

Shapiro-Wilk

Anderson-Darling



SHAPIRO-WILK



SHAPIRO-WILK

The Shapiro-Wilk normality test calculates a statistic based on the correlation between the data and the corresponding normal distribution; effective for sample size ($n \leq 50$).

Hypothesis

H_0 : Normal data

H_a : Non-normal data ($p\text{-value} < \alpha$)

Assumption

- Continuous data

syntax

```
from scipy import stats
```

```
w_stat, p_value = stats.shapiro(data)
```



EXERCISE

Perform Shapiro-Wilk normality test for the given dataset.

dataset

defects-30-sample.csv

Solution

```
from scipy import stats  
  
w_stat, p_value =  
stats.shapiro(df['Defects'])
```

Let $\alpha = 0.05$

Null Hypothesis

H_0 : Normal data

H_a : Non-normal data (p-value < 0.05)



ANDERSON-DARLING



ANDERSON-DARLING

The Anderson-Darling normality test provides a more sensitive test by giving weight to the tails of the distribution; effective for larger sample size ($n > 50$).

Hypothesis

$H_0: A^2 \leq \text{critical value}$ (Normal data)

$H_a: A^2 > \text{critical value}$ (Non-normal data)

Assumption

- Continuous data

Syntax

```
from scipy import stats

a2_stat, critical, alpha =
    stats.anderson(data)
```



EXERCISE

Perform Anderson-Darling normality test for the given dataset.

dataset

defects-count.csv

Solution

```
from scipy import stats  
  
a2_stat, critical, alpha =  
    stats.anderson(df['Defects'])
```

Let $\alpha = 0.05$

Null Hypothesis

$H_0: A^2 \leq \text{critical value}$ (Normal data)

Alternative Hypothesis

$H_a: A^2 > \text{critical value}$ (Non-normal data)



LABORATORY

